

1. Define Heredity, variations, genetics.

Ans: Heredity; Transmission of characters from 1 generation to other is called as heredity.

Variation: Difference between parent and offsprings and among the offspring of the same parent is called as variations.

Genetics: The branch of biology which deals with study of heredity and variation is called as genetics.

2. Enlist Mendel's 7 contrasting traits.

Ans:

Characteristic studied	Dominant character	Recessive character
Seed shape	round 	wrinkled 
Seed colour	yellow 	green 
Seed-coat colour	coloured 	white 
Pod shape	inflated 	constricted 
Pod colour	green 	yellow 
Flower position	axial 	terminal 
Stem length	long 	short 

Q3.Mendel selected pea plant for his experiment . Explain.

Ans: Mendel selected pea plant for the following reasons.

1. Annual plant with very short span of 3-4 months
2. Small herbaceous plant with many seeds
3. Self pollinating plant which can be cross fertilized artificially
4. Its seeds are viable
5. It does not exhibit intermediate characters.
6. Its flowers are large conspicuous easy for emasculation.

Marking scheme: any 4 points each point 0.5 marks hence 2 marks

4. State and explain Mendel's first law of Inheritance. 7M

Ans: Mendel's first law of inheritance is Law of Dominance

Law of Dominance: *In a cross between 2 organisms pure for any pair of contrasting characters, the character that appear in F₁ generation is called dominant and the one which is suppressed is called recessive.*

Explanation:

1. Mendel selected a variety of tall pea female parent and another dwarf male pea parent.
2. In F₁ generation he got all tall pea plants
3. On selfing he received 3 tall and 1 dwarf pea plant
4. The genotypic ratio is 1:2:1 with TT, Tt, tt genotype respectively

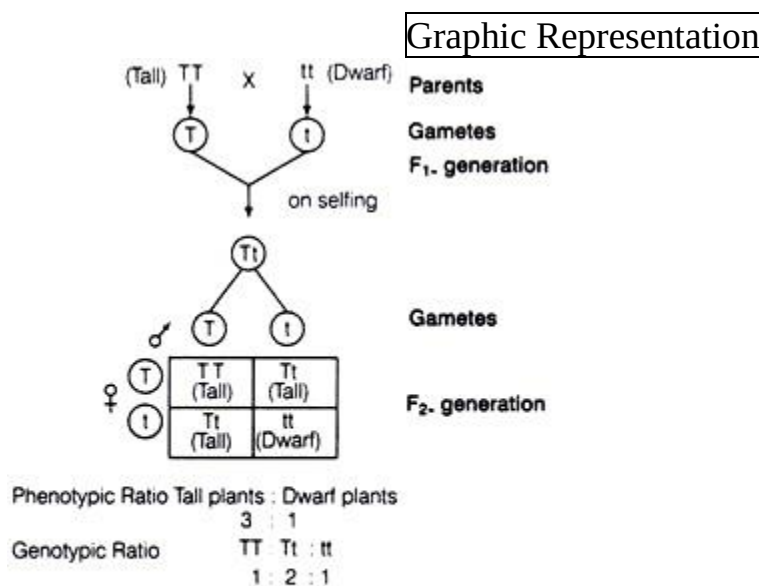


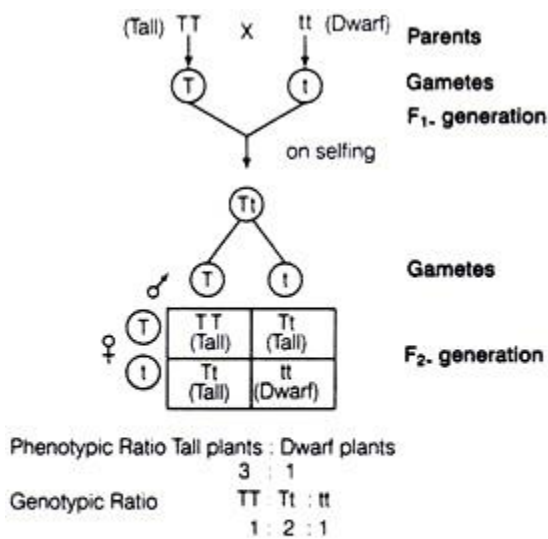
Fig. 5.2 A punnett square used to know a typical monohybrid cross conducted by Mendel between true-breeding tall plants and true-breeding dwarf plants

5. Explain Monohybrid cross with suitable example? 3M

Ans: Monohybrid Cross: *a cross between 2 pure parents in which inheritance pattern of only 1 pair of contrasting characters is studied is called monohybrid cross.*

Explanation:

1. Mendel selected a variety of tall pea female parent and another dwarf male pea parent.
6. In F₁ generation he got all tall pea plants
7. On selfing he received 3 tall and 1 dwarf pea plant
8. The genotypic ratio is 1:2:1 with TT, Tt, tt genotype respectively



On Selfing

Fig. 5.2 A punnett square used to know a typical monohybrid cross conducted by Mendel between true-breeding tall plants and true-breeding dwarf plants

9.

6.State and explain Mendel's second law of Inheritance.

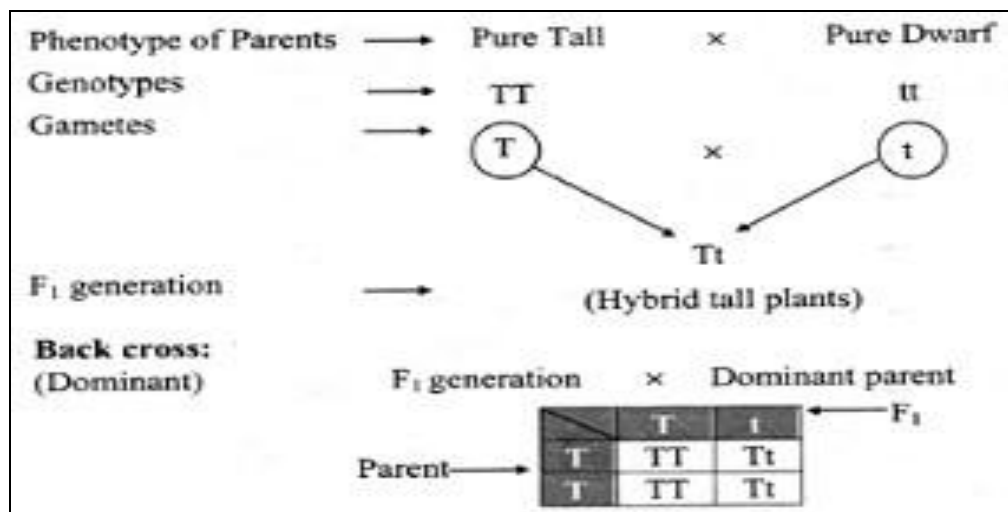
Ans: Mendel's second law of inheritance is Law of segregation

Statement of law:

Members of allelic pair in a hybrid remain together without mixing with each other and separate or segregate during gamete formation. Thus gametes receive only 1 of the 2 factors and are pure for the given trait. Therefore this is known as law of purity of gametes.

Explanation

1. Law of segregation can be explained with the help of test cross.
2. **Test cross:** *it can be defined as a cross between individual with unknown genotype for a particular trait with a recessive plant for that trait.*
3. In this cross heterologous trait e.g. heterologous tall pea plant is crossed with homozygous dwarf pea plant.
4. Reappearance of recessive trait in F₂ generation proves the law.
5. Phenotypic ratio is 2:2
6. Genotypic ratio is 2:2



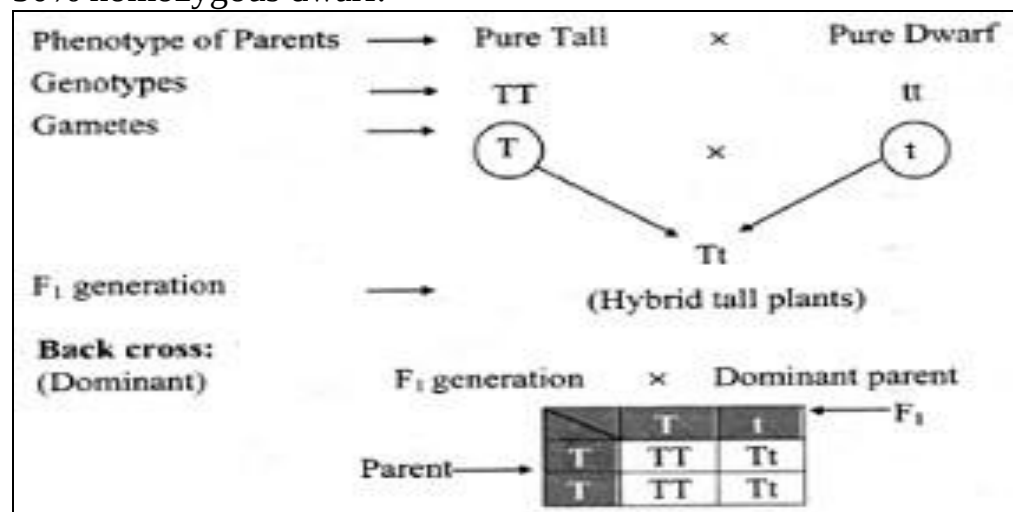
Test cross	Heterozygous F_1 Tall	$\times$	Homozygous Dwarf parent
Genotypes	(Tt)		(tt)
Types of gametes	(T) (t)		(t)
	(T) (t)		
Test cross progeny	(t)	Tt Tall	tt Dwarf
		1 Tall : 1 Dwarf	50% 50%

Phenotypic ratio:	2	:	2
	Heterozygous tall		homozygous Dwarf
Genotypic ratio	2	:	2
	Tt		tt

7. All test cross are back cross but all back cross are not test cross. Give reason. (2m)

Ans:

1. **Test cross:** It can be defined as a cross between individual with unknown genotype for a particular trait with a recessive plant for that trait.
2. **Back cross:** cross between F_1 generation and any of its homozygous parent is called as back cross.
3. Cross of F_1 generation and dominant parent will produce all tall offspring with 50% homozygous tall and 50% heterozygous tall.
4. But cross of F_1 generation with recessive parent will create 50% tall and 50% homozygous dwarf.



Test cross	Heterozygous F_1 Tall	X	Homozygous Dwarf parent	
Genotypes	(Tt)		(tt)	
Types of gametes	(T) (t)		(t)	
	(T)		(t)	
Test cross progeny	(t)	Tt Tall	tt Dwarf	1 Tall : 1 Dwarf 50% 50%

Q8. A heterozygous tall plant of pea is crossed with a dwarf plant of pea. Caculate phenotypic ratio of the progeny. (3M)

Ans:

P generation: heterozygous tall X Dwarf

Phenotype: Tall Dwarf

Genotype: Tt tt

Gamete T t t t

F_1 generation: Tt

On crossing F_1 with recessive parent i.e

Tt X tt

	T	t
		
t	Tt tall	tt dwarf
t	Tt tall	tt dwarf

Phenotypic ratio: 2 : 2
Heterozygous tall homozygous Dwarf

Genotypic ratio 2 : 2
Tt tt

Marking Scheme: Genotype of parents= 0.5m, gametes of parents= 0.5 m, checker board=1m, phenotypic ratio= 1m

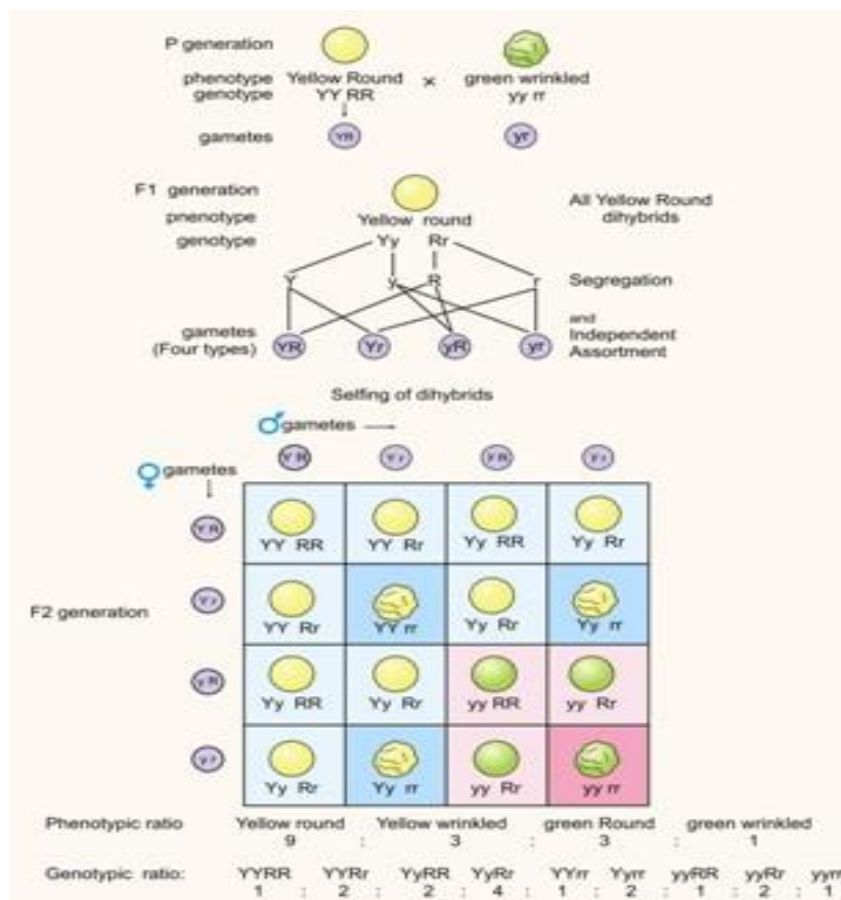
Q9. State and explain Law of Independent Assortment with a suitable example.(7M)

Ans: Law of independent assortment

When the two homozygous individuals differ from each other, in 2 or more pairs of contrasting characters or genes are crossed, then the inheritance of one pair of character is independent of the other character.

Explanation:

1. Mendel crossed a homozygous pea plant having yellow round seeds (YYRR) with a plant having green wrinkled seeds (yyrr)
2. The plants of F₁ generation are all heterozygous. (YyRr).
3. The 4 types of alleles are separated into 4 types of gametes. (YR), (Yr), (yR), (yr)
4. In addition to parental combination of yellow round and green wrinkled, 2 new combinations of yellow wrinkled and green round seeds are produced.
5. The appearance of new combination in F₁ generation proves the law.
6. They showed that each pair of contrasting character behaves independently and has no permanent association with a particular character.
7. **Graphic representation of dihybrid cross:**



Marking Scheme: Statement of law=1m, explanation=2m, graphic representation=3M, Ratios=1M

Q10. Explain dihybrid cross with the suitable example.(7M)

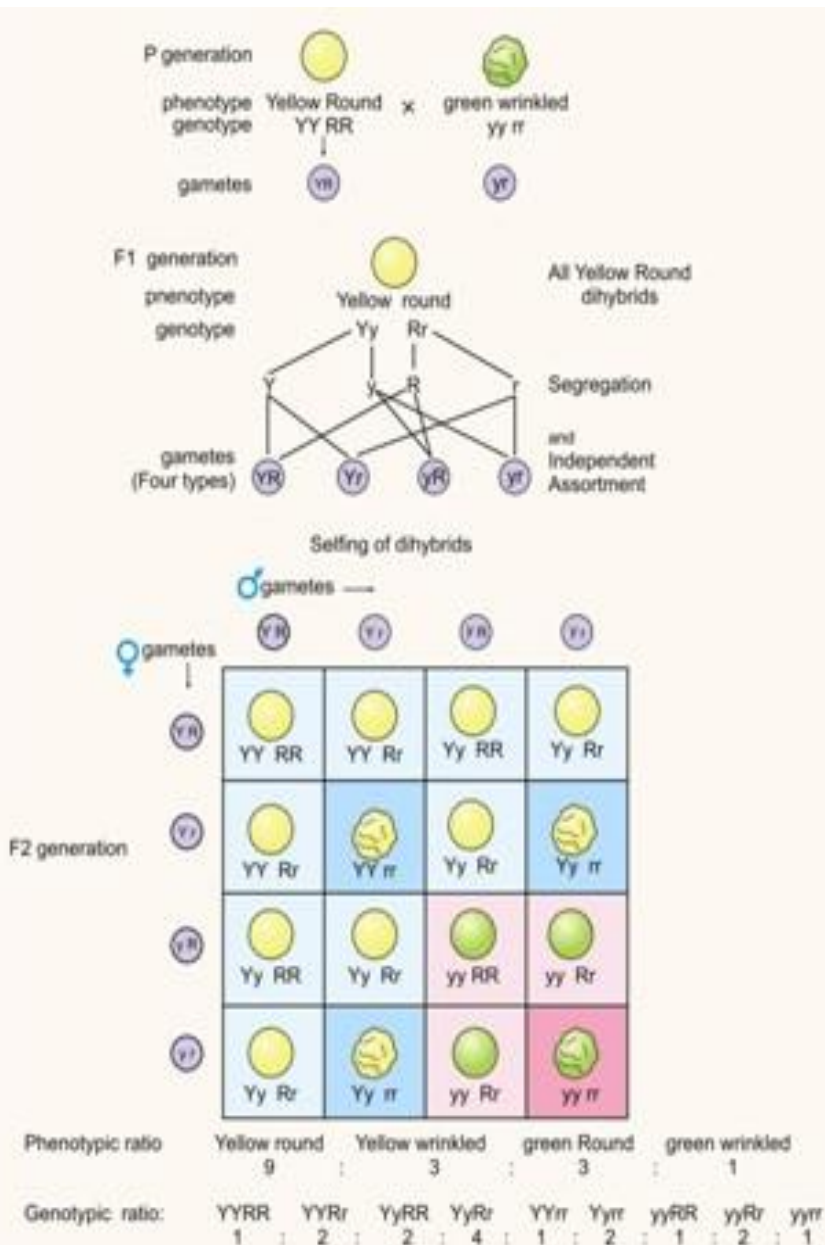
ANS: Dihybrid cross

Def: A cross between 2 pure (homozygous) parents in which the inheritance pattern of 2 pairs of contrasting characters is considered simultaneously is called dihybrid cross.

Explanation:

1. Mendel selected a variety of pea plants having yellow and round seeds as female parent and another variety have green and wrinkled seeds as male parent.
2. He obtained pure line by selfing for 3 generations.

Graphic representation of dihybrid cross:



Marking Scheme: definition= 1m, explanation=1m, cross till F1 generation=1 marks, F2 generation=3 m, Phenotypic and genotypic ratios= 1m.

Q 11. Describe the cross between homozygous tall, round seeded pea plant and a dwarf, wrinkled seeded pea plant. What will be the type of progeny in F2 generation of this cross and in what proportion will it be/ name and state the law which is explained by this example.

- Ans:**
1. Consider a cross between homozygous tall and round seeded pea plant (TTRR) and a dwarf and wrinkled seeded pea plants. (ttrr)
 3. The plants of F1 are heterozygous for both characters with the genotype TtRr.
 4. The F1 hybrid is allowed to self pollinate
 5. The gametes which are produced is TR, Tr, tR, tr
 6. In addition to parental combination of tall wrinkled and dwarf round, are produced.

Independent Assortment

Parents: Tall, Round (TtRr) x Tall, Round (TtRr)
Gametes: TR; Tr; tR; tr

Gametes	TR	Tr	tR	tr	9:3:3:1
TR	TTRR	TTRr	TtRR	TtRr	9/16 Tall, Round
Tr	TTRr	TTrr	TtRr	Ttrr	3/16 Tall, Wrinkled
tR	TtRR	TtRr	ttRR	ttRr	3/16 Dwarf, Round
tr	TtRr	Ttrr	ttRr	ttrr	1/16 Dwarf, Wrinkled

NOTE: The tall, wrinkled (TTrr and Ttrr genotypes), dwarf, round (ttRR and ttRr genotypes) and dwarf, wrinkled (ttrr genotypes) progeny are called recombinants because they differ to the parental genotypes and phenotypes

:

Phenotypic ratio: 9:3:3:1

Genotypic Ratio: 1:2:2:4:1:2:1:2:1

The law which is explained in this example is law of Independent Assortment.

Statement of Law

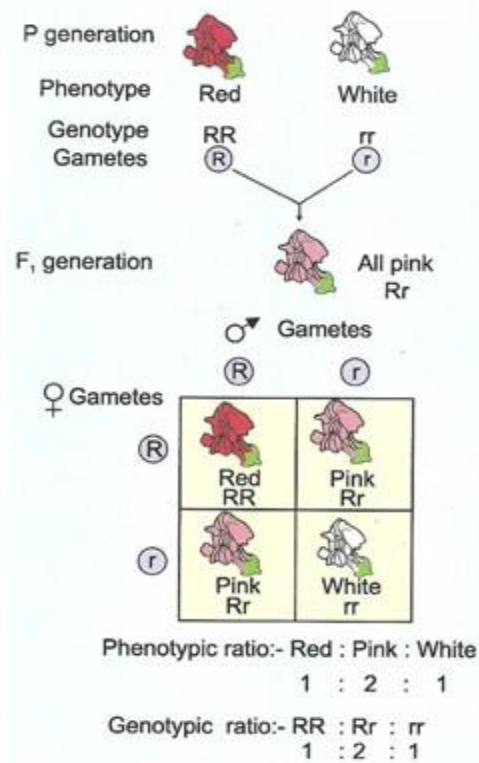
“ When 2 homozygous individuals differ from each other, in 2 / more pairs of contrasting characters or genes are crossed then the inheritance of 1 pair of character is independent of the other pair of the character.”

Marking scheme: Dihybrid cross=1m, Explanation=1m, F2 generation= 2.5m, types of progeny 1 m, name of law=0.5 m, statement of law =1m

Q12. In in-complete dominance and co-dominance , genotypic and phenotypic ratios are identical. Explain how co-dominance differs from incomplete dominance in phenotypic nature of their hybrids. (3M)

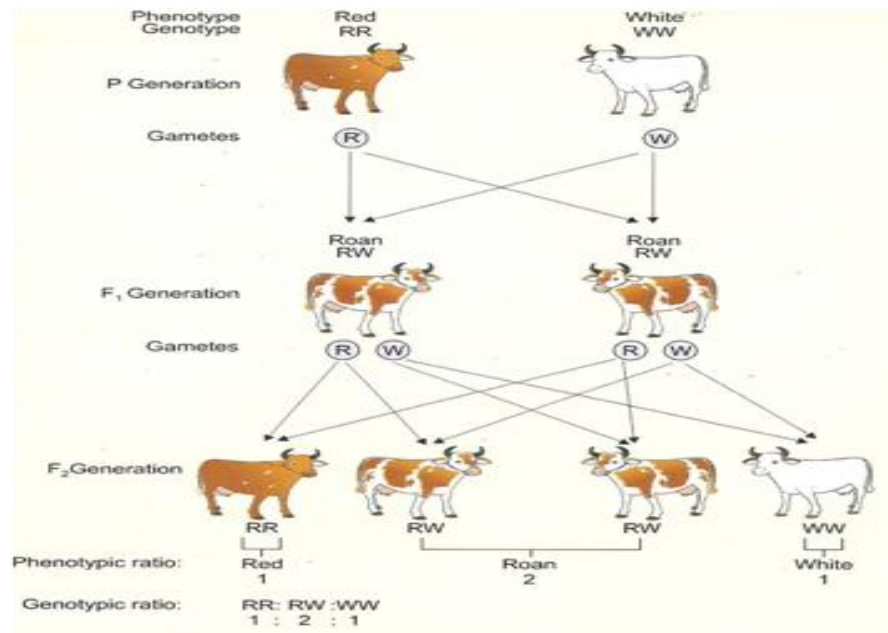
Ans: Incomplete dominance :

1. In incomplete dominance both the genes of an allelomorphic pair express themselves partially .
2. One gene cannot suppress the expression of the other completely.
3. There is an intermediate expression in F₁ hybrid.



Co-dominance

1. In codominance both the genes of an allelomorphic pair express themselves equally in F₁ hybrids.
2. These alleles express themselves independently even if present together in hybrids.



3.

Difference : In both incomplete dominance and codominance, phenotypic and genotypic ratios are identical, but in a case of incomplete dominance the phenotype of hybrids are intermediate between the phenotypes of the parents, while in codominance both genes are expressed equally.

Marking Scheme: Codominance: explanation and definition 1 mark, Incomplete dominance: explanation and definition 1 mark, difference between them 1 mark += 3 mark

Q13. A pea plant pure for yellow seed colour is crossed with a pea plant for green seed colour. In F₁ generation all pea plants were with yellow seeds. Which law of Mendel is applicable. (1mark)

Ans: Law of Dominance or Mendel's First law of Inheritance is applicable.

Q14. Write a note on multiple alleles considering example of human blood groups.3M

Ans:Multiple Alleles

“More than 2 alternative form of gene in a population occupying same locus on a chromosome or its homologue are known as multiple alleles.”

Explanation:

- The gene I controls ABO blood groups
- It Has 3 alleles I^A, I^B, i
- Allele I^A and I^B produce different type of sugar/ antigen
- Allele i do not produce any antigen/ sugar.
- The blood group chart is as follows

Blood Type	Antigens	Antibodies	Can Give Blood (RBCs) to	Can Receive Blood (RBCs) From
AB	A and B	None	AB	AB, A, B, O
A	A	B	A and AB	A and O
B	B	A	B and AB	B and O
O	None	A and B	AB, A, B, O	O

Q15. What is test cross? 1M.

Ans: When F1 hybrid is back crossed with its recessive parent, it is called test cross.

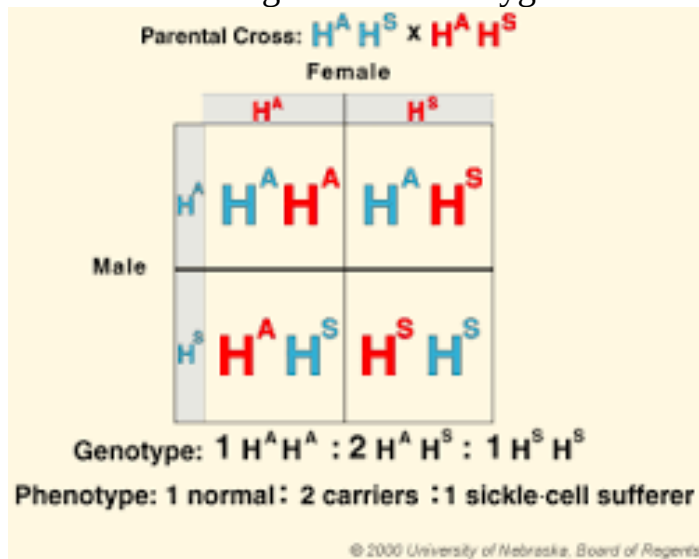
Q16. Why the ratio in pleiotropy is 2:1? Explain it with example.(3M)

Ans:Pleiotropy: “ When a single gene controls 2 or more different traits, it is called pleiotropic genes and this phenomenon is called as pleiotropy.

Its ratio is 2:1 instead of 3:1

Explanation:

1. According to Mendel's principle of unit character 1 gene controls 1 character, but sometimes single gene produces 2 related or unrelated phenotypic expressions.
2. **Example: Sickle cell Anaemia**
Sickle cell anaemia- Hb^S (Recessive)
Normal healthy gene- Hb^A (Dominant)
Carrier – $Hb^A Hb^S$ (Heterozygous)
3. The carrier heterozygotes show sign of mild anaemia as their RBC's become sickle shaped in oxygen deficiency. They are said to have sickle cell trait but are considered normal.
4. Homozygotes ($Hb^A Hb^S$) develop severe anemia and die. Hence Sickle cell anemia is lethal.
5. Marriage between 2 carriers will lead to produce 1 normal, 2 carriers and 1 anemic child which dies.
6. Hence the ratio is 2:1
7. Therefore marriages of 2 heterozygote carriers should be avoided.



Marking scheme: Description=1.5m, chart-1.5 m

1. **Law of dominance is not universally applicable. Give Reason**
 2. The law of dominance is not universally applicable because as in intra allelic interaction neither gene is dominant nor recessive.
 3. Sometimes both gene have equal potential to express as found in incomplete dominance and co dominance. it modifies the mendelian ratio of 3:1 to 1:2:1.
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Chromosomal Basis of Inheritance

Q2. Explain structure of chromosome in detail.(3M) (Diagram 1m, all highlighted words explanation 2m)

Ans: 1. A metaphasic chromosome has 2 identical halves called sister chromatids.

2. Each chromatid is in turn made up of sub-chromatids called **chromonemata**.

3. The chromatids lie side by side & are held together at one point called the **centromere**.

4. The centromere is also called the **primary constriction**.

5. Secondary constrictions are also present.

6. The part of the chromosome beyond the nucleolar organiser is short, spherical & is called **satellite**.

7. The tip of the chromosome is called **telomere**.

8. The surface of the chromosome bears number of small swellings called **chromomeres**.

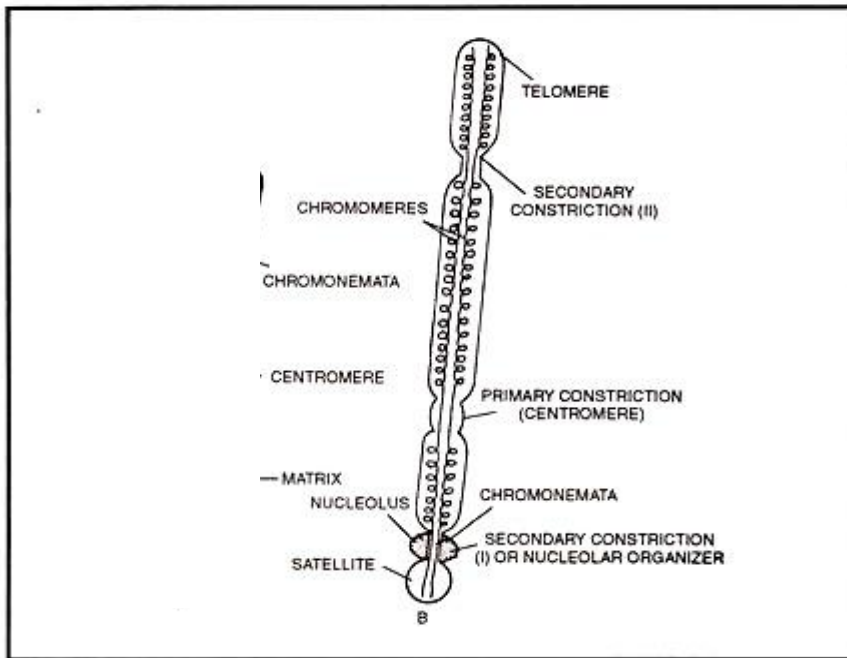


Fig. 49. Structure of chromosome (in metaphase stage)

Q3. State and explain types of chromosomes.(3M)

Ans: METACENTRIC CHROMOSOMES

Metacentric chromosomes have the centromere in the center, such that both sections are of equal length..It appears V shaped during anaphase.

SUBMETACENTRIC CHROMOSOMES

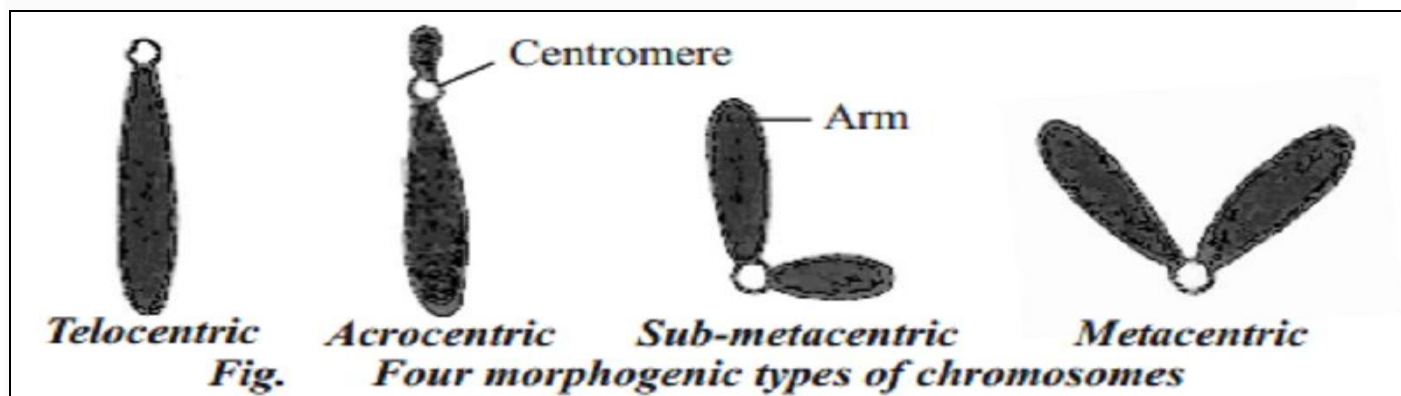
Submetacentric chromosomes have the centromere slightly offset from the center leading to a slight asymmetry in the length of the two sections.. It appears L shaped during anaphase.

ACROCENTRIC CHROMOSOMES

Acrocentric chromosomes have a centromere which is severely offset from the center leading to one very long and one very short section.. It appears J shaped during anaphase.

TELOCENTRIC CHROMOSOMES

Telocentric chromosomes have the centromere at the very end of the chromosome. Humans do not possess telocentric chromosomes but they are found in other species such as mice. It appears rod shaped during anaphase.



Q4. Distinguish between X and Y chromosome. (2m) (any 4 points)

Sr No	P.O.D	X- Chromosome	Y-Chromosome
1.	Length	It is longer than Y chromosome	It is shorter than X chromosome
2.	Euchromatin	Large amount of euchromatin present	Small amount of euchromatin.
3.	Heterochromatin	Small amount of heterochromatin	Large amount of heterochromatin present
4.	Genetically active	As more number of genetically active genes are present hence it is more genetically active	As less number of genetically active genes are present hence it is less genetically active
5.	Genes	Non homologous region of X chromosome contains X linked genes.	Non homologous region of Y- chromosome contains Y- linked genes.

Q5. What is sex linkage? Explain the inheritance of colorblindness & hemophilia with the suitable example.(7M)

Ans: Sex linkage: Transmission of body characters from parents to offspring along with sex chromosome is called sex inheritance or sex linkage.

A] Inheritance of color blindness:

1. It is a sex linked disease in which person cannot distinguish between red & green color. Both the color appears grey to him.
2. It is caused by recessive gene, which prevents the proper formation of color sensitive cells in the retina necessary for distinction of red & green.
3. The genes for normal vision (dominant) & colorblindness (recessive) are located on non-homologous region of X- chromosome. Their alleles are absent in Y- chromosome.
4. If gene for normal vision is represented by X^C & gene for color-blindness is represented by X^c , then the genotype of different individuals can be represented as follows:-

Sex	Normal	Colourblind	Carrier
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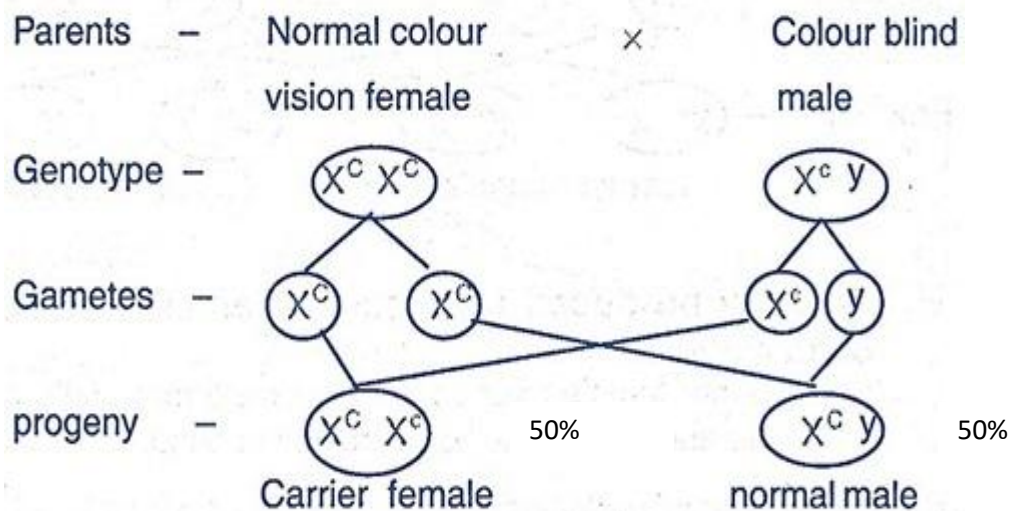
Male	$X^C Y$	$X^c Y$	-
Female	$X^C X^C$	$X^c X^c$	$X^c X^C$

Explanation :

The inheritance of colourblindness can be studied by following examples:

1. If a colourblindman ($X^c Y$) marries a female with normal vision ($X^C X^C$) all the offsprings will have normal vision. The sons will have normal vision and daughters will be all carriers.
2. If a carrier female ($X^C X^c$) marries a male with normal vision then all daughters will have normal vision of which half will be carriers.
3. Half the sons will be colourblind and half of them will have normal vision
4. Hence it is clear that the colourblind father transmits the disease to his grandson through his carrier daughter
5. This inheritance from father to grandson through his daughter is called **CRISS CROSS INHERITANCE**.

Chart



PROGENY PRODUCED FROM NORMAL COLOUR VISION FEMALE AND COLOUR BLIND MALE

Phenotypic Ratio of F_1 Generation: $X^C X^c$: $X^C Y$

Carrier female Normal Male

50% 50%

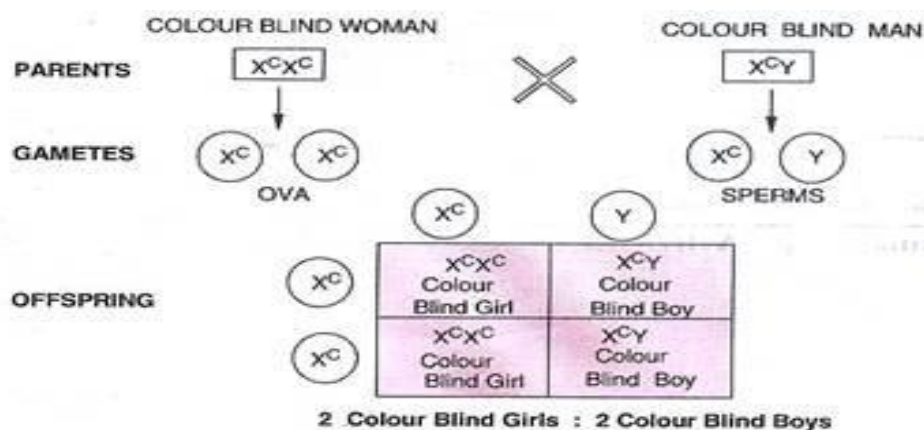


Fig. 5.55. Progeny of a colour blind woman and a colour blind man.

Phenotypic Ratio of F_2 Generation: $X^C X^C$: $X^C X^c$: $X^c Y$: $X^C Y$

male	Normal female	Carrier female	Normal male	Colourblind
	25%	25%	25%	25%

B] Inheritance of Haemophilia

1. Haemophilia is a hereditary disease in which blood fails to clot or clots very slowly.
2. Person who carries the recessive gene for haemophilia has deficiency of clotting factors in blood. So minor injuries causes continuous bleeding. Hence it is also called as bleeder's disease.
3. The genes for normal clotting (dominant) and haemophilic (recessive) are located on non homologous region of X chromosome. But their alleles are absent in Y-chromosomes
4. Genes for normal clotting of blood is represented by X^H & gene for haemophilic by X^h .
The genotype of different individuals can be represented as follows:

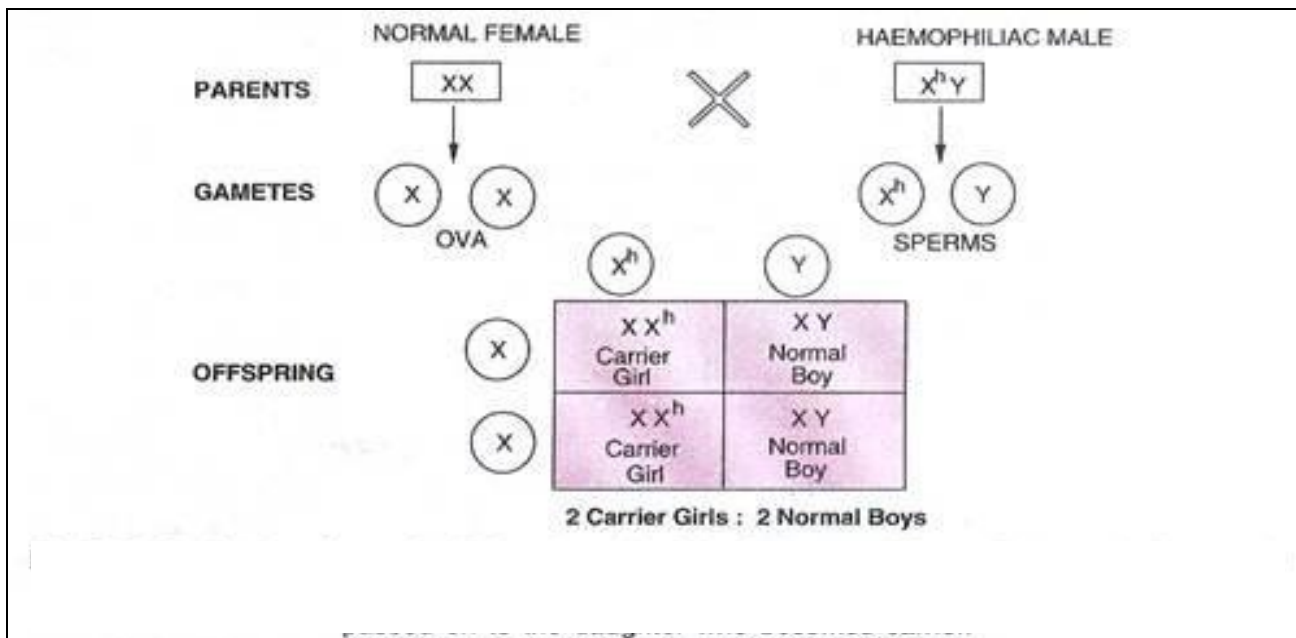
Sex	Normal	Colourblind	Carrier
Male	$X^H Y$	$X^h Y$	-
Female	$X^H X^H$	$X^h X^h$	$X^h X^H$

Explanation :

The inheritance of haemophilia can be studied by following examples:

1. If haemophilic male ($X^h Y$) marries a normal female ($X^H X^H$) then all the offsprings will have normal clotting of blood. The son's will have normal blood clotting but the daughters will be carriers for the disease. The carriers have normal clotting of blood.
2. If carrier female ($X^H X^h$) marries a Male with normal clotting of blood ($X^H Y$), then all daughters will have normal clotting of blood but half of them will be carriers of the disease.
3. Half the son will be haemophilic & remaining half will have normal clotting of blood.
4. Haemophilia also shows **CRISS CROSS INHERITANCE**.
5. Hence, father transmits the disease to his grandson through his carrier daughter

Charts:

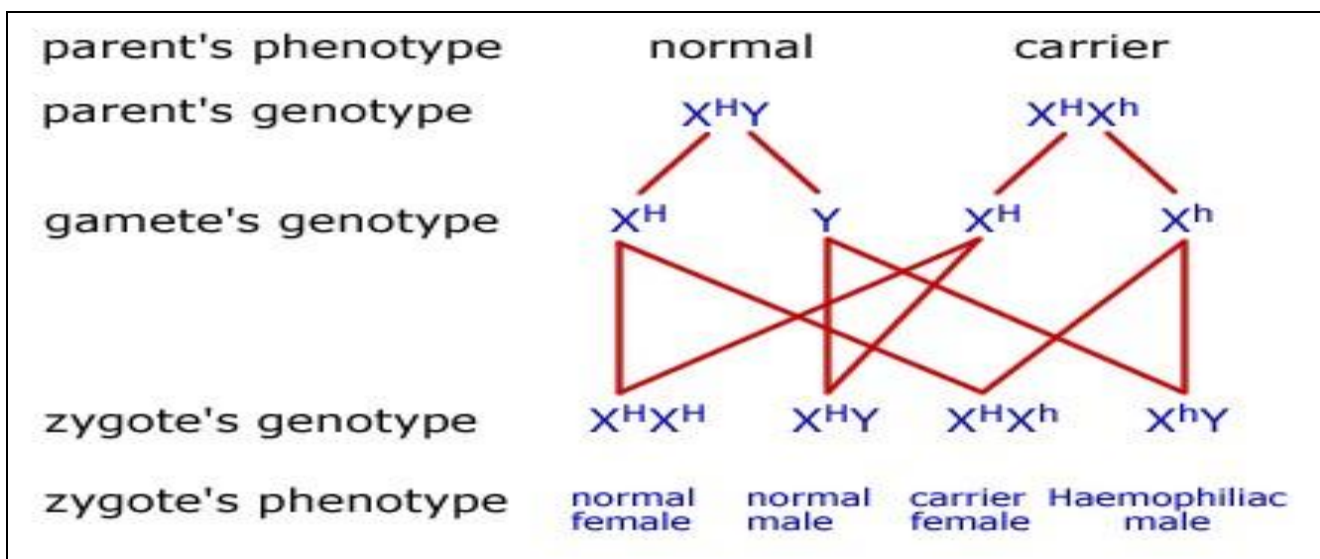


Phenotypic Ratio of F_1 Generation: X^hXH : X^HY

Carrier female Normal Male

50%

50%



Phenotypic Ratio of F_2 Generation: X^HX^H : X^HX^h : X^HY : X^hY

Normal female

Carrier female

Normal male

haemophilic

male

25%

25%

25%

25%

(marking scheme: Def of sex linkage 1m, Colourblindness:1m, explanation of eg:1m, colourblindness chart:1m, haemophilia:1m, explanation of eg:1m, haemophilia chart 1m)

Q6. Why do sex linked traits appear more in males than in females. (1M)

Ans: In males, there is only 1 X-chromosome, hence recessive gene get a chance to express itself.

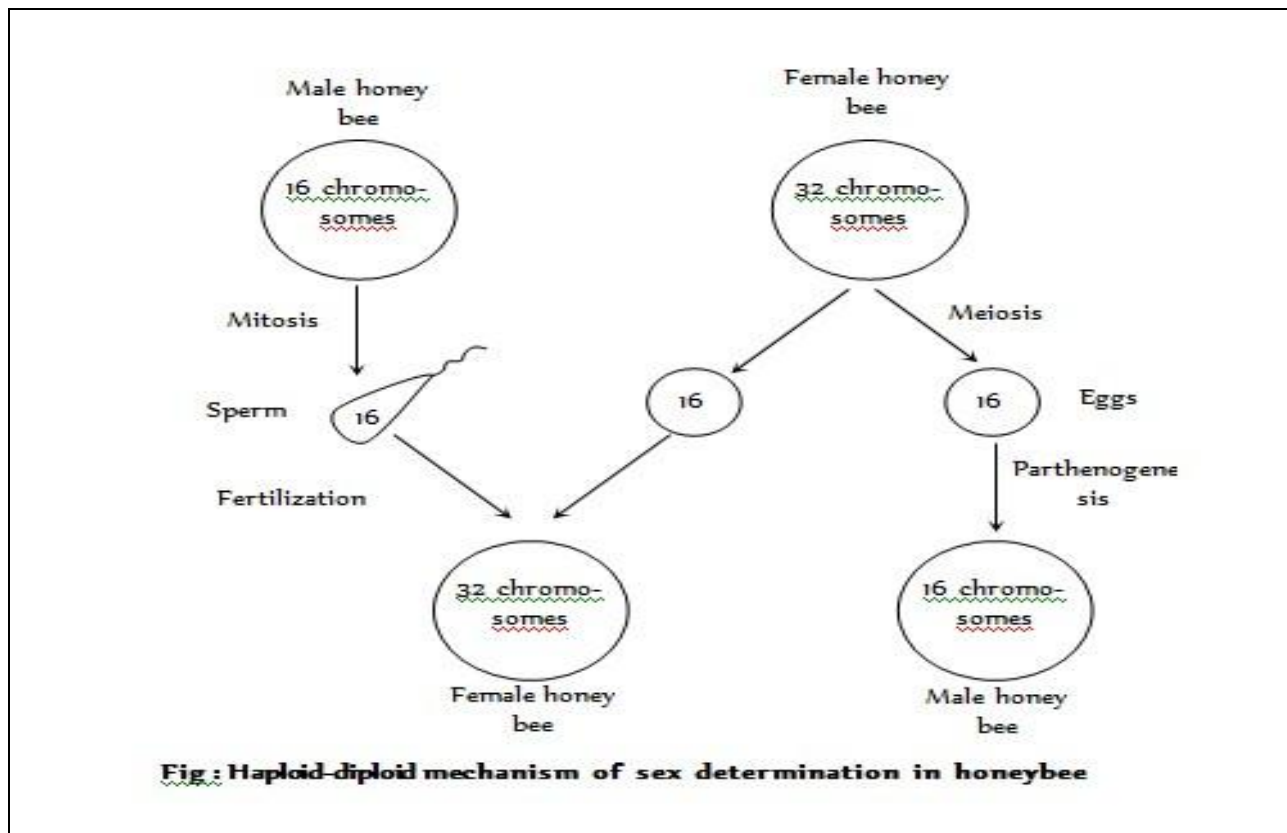
Q7. With the help of chart explain the method of sex determination in honeybees.(3M) /

Drone of honeybees show haploid number of chromosomes. Illustrate. (chart:1.5m, explanation :1.5m)

Ans: Sex determination in honeybees.

1. In honeybees, sex is determined by the number of sets of chromosomes received by an individual. Such type of sex determination is called haploid-diploid sex determination system.
2. The fertilized egg develops as a female offspring (may be queen or worker bee). It shows diploid number of chromosomes i.e $2n=32$.
3. An unfertilized egg develops as a male (drone) by means of parthenogenesis. Thus the drones have haploid $n=16$ number of chromosomes. The drones produce sperm by mitosis.

Chart;



Q8. Describe the structure of sex chromosome. (3M) (1.5m: explanation, 1.5m: diagram)

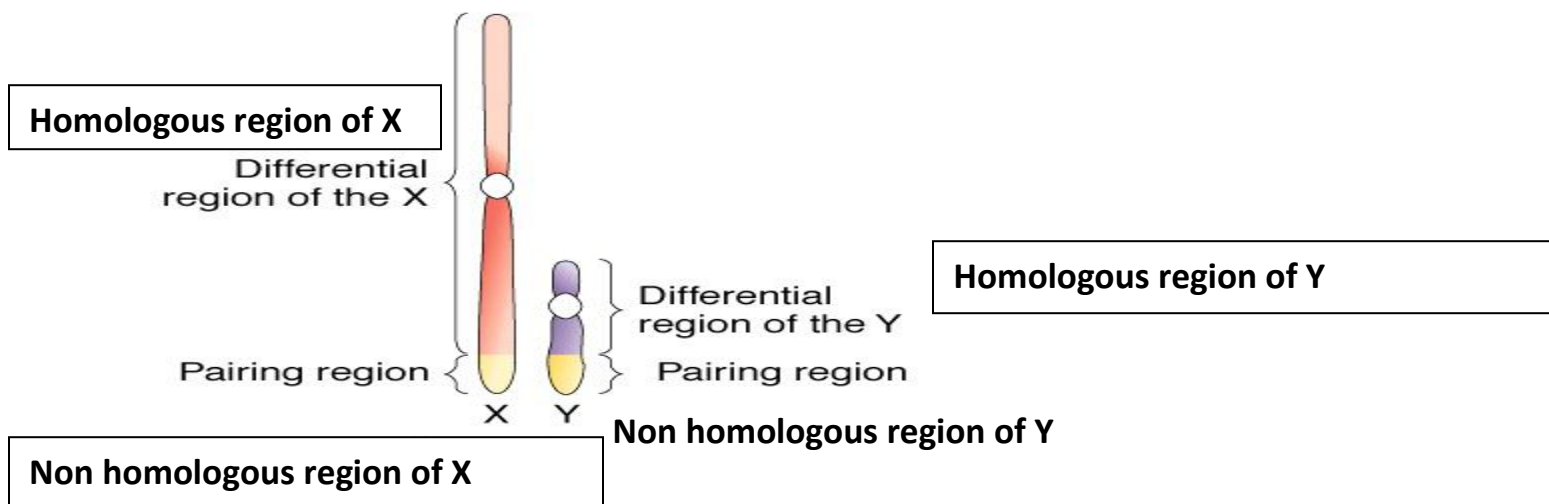
Ans: X- chromosome:

1. It is longer & contains large amount of euchromatin & small amount of heterochromatin
2. It is common in males & females.
3. Its non homologous part contains X-linked genes that show criss cross inheritance.

Y- Chromosome

1. It is shorter than X- chromosome.
2. It contains small amount of euchromatin & large amount of heterochromatin

3. Its non homologous part contains Y-linked genes that show straight inheritance.



Q9. Give any 2 sex linked traits.

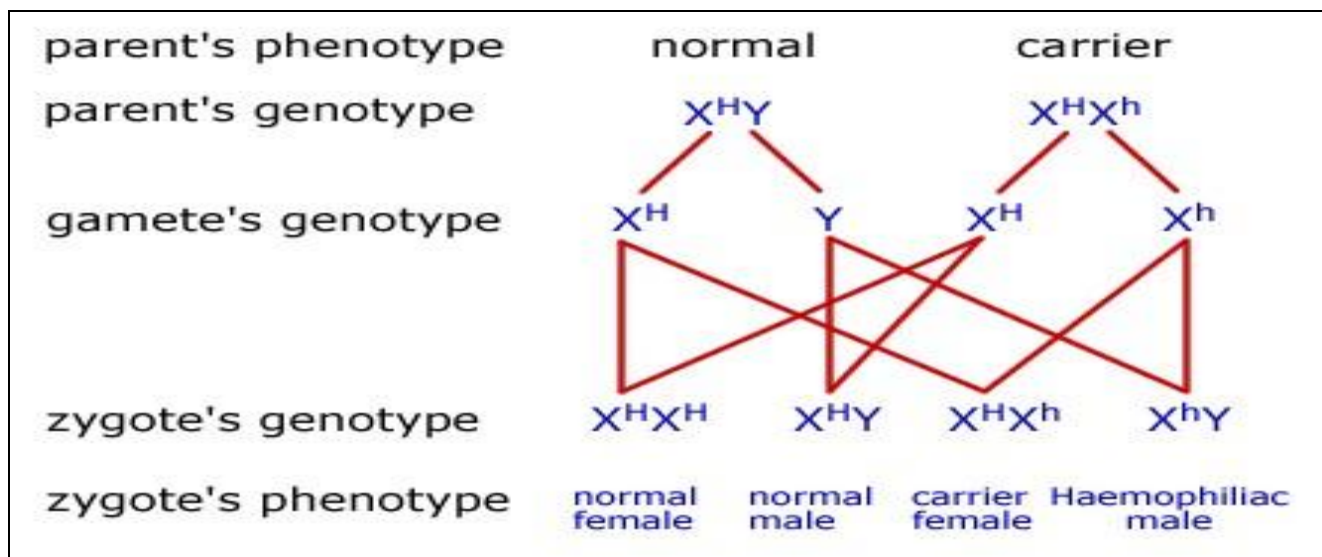
Ans: X- linked diseases are:-

- a) Colourblindness b) Haemophilia c) Myopia d) Muscular dystrophy e) night blindness

Q10. What will be the phenotype of progeny, if a carrier hemophilic female marries a normal male? Explain.

Ans: if a carrier hemophilic female marries a normal male then:

1. If carrier female ($X^H X^h$) marries a Male with normal clotting of blood ($X^H Y$), then all daughters will have normal clotting of blood but half of them will be carriers of the disease.
2. Half the son will be haemophilic & remaining half will have normal clotting of blood.



Phenotypic Ratio : $X^H X^H$: $X^H X^h$: $X^H Y$: $X^h Y$

Normal female

Carrier female

Normal male

haemophilic male

25%

25%

25%

25%

Q11. What is a criss cross inheritance. (1M)

Ans: The inheritance of characters from the father to his grandson through his carrier daughter is called Criss Cross inheritance.

Q12. Sketch & label structure of chromosome. (2M) (any 4 labels)

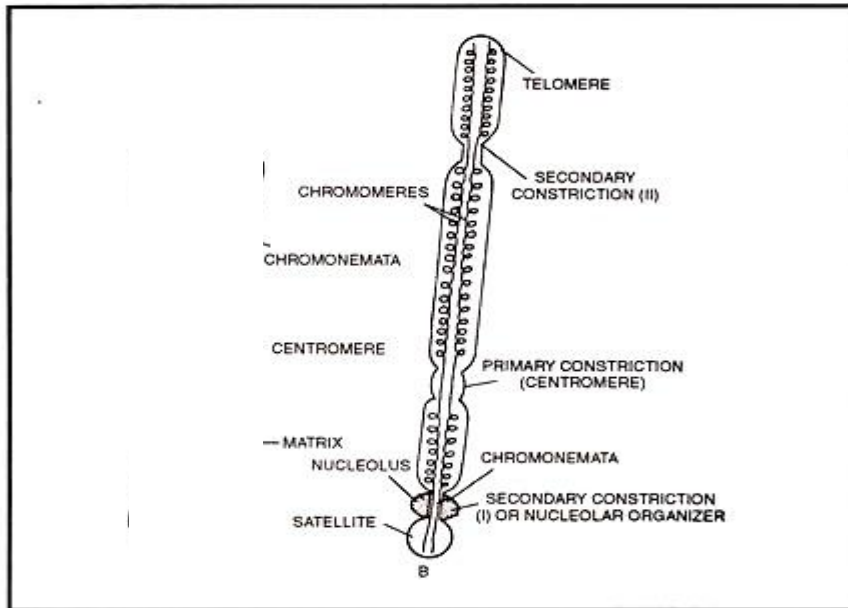


Fig. 49. Structure of chromosome (in metaphase stage)

Ans:

Q14. Give the genotype of Turner's syndrome.

Ans: 44A+ XO is the genotype of Turner's syndrome

Q15. Explain sex chromosomal abnormalities

Ans: A] Turner's syndrome

1. It was described by HH Turner.
2. Person with Turner's syndrome have 45 chromosomes with only 1 X sex chromosome, such a genotype is represented as (44+ XO)

Symptoms of Turner syndrome are:

- Phenotype is female
- short stature
- infertility,
- webbed neck,
- No secondary sexual characters.

B] Klinefelter's Syndrome

1. It is caused due to an extra chromosome i.e. extra X chromosome. So the genotype is XXY so they are sometimes described as feminized males.
2. The sufferer has 47 chromosomes instead of 46

Symptoms:

1. Patients are tall, thin sterile
2. Normal intelligence

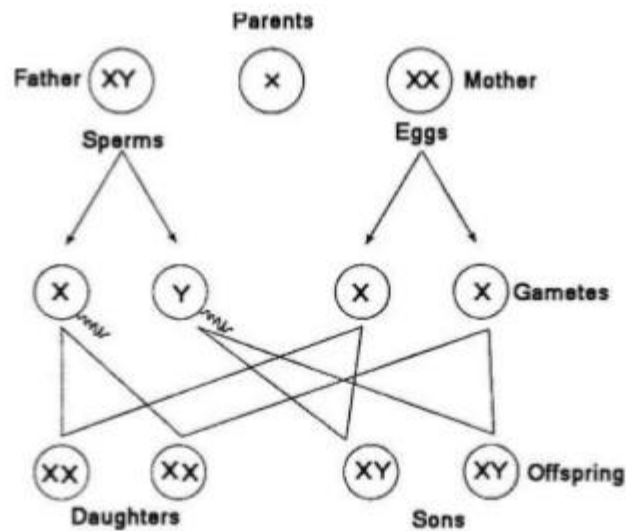
3. Spermatogenesis is absent

Q13. Explain the mechanism of sex determination in human being.

Determination of sex in the human being

Sex determination is due to sex chromosomes. Therefore, it is a genetic mechanism.

In human beings the female possesses 44 autosomes and a pair of homomorphic sex chromosomes, i.e. XX, while the male possesses 44 autosomes and a pair of heteromorphic sex chromosomes, i.e. XY.



Thus all normal males are 44 + XY and all normal females are 44 + XX.

At the time of gamete formation, the meiosis occurs.

The sperms produced by males are of two types, one containing 22 autosomes and one X chromosome and other containing 22 autosomes and one Y chromosome.

The eggs produced by females are of only one type with 22 autosomes and one X chromosome.

When X bearing sperm fertilizes the egg, the female child is conceived having the combination 44 + XX.

When Y bearing sperm fertilizes the egg the male child is conceived having the combination of 44 + XY.

Hence father is responsible for the sex of the child and not the mother.

Q16. Distinguish Between Crossing over & linkage

Ans

Difference between crossing over and linkage

Crossing over	Linkage
1. It leads to separation of linked genes	1. It keeps the genes together
2. It involves exchange of segments between non-sister chromatids of homologous chromosomes	2. It involves individual chromosomes
3. The frequency of crossing over can never exceed 50 %	3. The number of linkage groups can never be more than haploid chromosome number
4. It increases variability by forming new gene combinations	4. It reduces variability

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